

Ashmit Dutta

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science, Computer Engineering

Aug 2023 – May 2027

- **Relevant coursework:** Safe Autonomy (ECE 484), GPU Programming (ECE 408), Deep Learning (CS 444).
- Co-author on an [ICML 2025 workshop paper](#) on finetuning Chain of Thought in LLMs.

EXPERIENCE

Nablon AI

Jun 2026 – Present

AI Engineering Intern

Remote — San Francisco, CA

- Building **Sentinel**, an automated LLM-evaluation framework for an enterprise automotive client: a **5-tier scanner pipeline** (rules, embeddings, classifiers, SLMs, LLM-as-judge) flagging jailbreaks, PII, toxicity, and hallucinations.

Assured and Trustworthy AI Research Lab – University of Illinois

Dec 2025 – Present

ML Research Intern (Prof. Huan Zhang)

Urbana, IL

- Implemented the **Softmax** operator in [auto.LiRPA](#), an open-source neural-network verification library (VNN-COMP winner); wrote C++ patches **cutting peak GPU memory 44%** and Python sanity-checking pipelines across 16+ benchmarks (ResNets, ViTs, YOLO, control nets).

John Deere

Jan 2025 – Jan 2026

Software Engineer Intern

Champaign, IL

- Shipped a React + Spring Boot data platform **to production** for **500+ users**, built on REST APIs and CI/CD over PostgreSQL/DB2 handling **millions of records**.
- Designed a multithreaded job scheduler **cutting DB search time 3x**; added AWS CloudWatch observability.

University of Chicago Data Science Institute — DSI Summer Lab

Jun 2024 – Aug 2024

ML Research Intern (Fermilab collaboration)

Chicago, IL

- Built a reusable multi-modal [Graph Neural Network library](#) (PyTorch Geometric) for particle-track reconstruction from **20,000+ 3D detector events** on a Slurm GPU cluster (Fermilab Exa.TrkX).

PROJECTS

[Autonomous Drone Racing](#) | ROS 2, Python, C++, PyTorch

Oct 2025 – Dec 2025

- Built a ROS 2 autonomy stack (min-snap planner + geometric controller) for a Crazyflie 2.1; tuned gains via Bayesian optimization, **outperforming 3 published baselines** on real hardware.
- Trained a **YOLOv8 gate-segmentation model** on **NeRF-rendered 3D data**, enabling vision-based localization through race gates.
- Analyzed flight logs (Rosbags) to debug pose-tracking error and validate localization accuracy on hardware.

[Multi-Modal Video Search Engine](#) | Python, Modal, FastAPI

Feb 2026

- **Winner** — **Modal Inference + Bright Data AI tracks at Stanford TreeHacks 2026** (out of 380 teams).
- Built a multi-modal embedding pipeline fusing **vision and audio** for semantic video retrieval on cloud GPU inference, indexing 24 hr+ of video.

[GPU-Accelerated Deep Learning Inference \(LeNet-5\)](#) | C++, CUDA

Feb 2026 – Apr 2026

- Wrote shape-specialized implicit-GEMM CUDA kernels and a WMMA Tensor Core conv for a **6.6x speedup** (12.5 ms inference at batch 10,000); **placed 5/130 in the ECE 408 competition**.

LEADERSHIP, PUBLICATIONS & AWARDS

Co-Founder & Lead Organizer, [Online Physics Olympiad \(OPhO\)](#): founded the largest student-run international physics competition (20k+ community); led website development and a team of 20+ students.

Awards & Leadership:

- *2nd Place, Bradley Mottier Innovation Challenge*, UIUC ISE (\$2,000) — [computer-vision AR glasses for ASL translation](#).
- *Executive Board Member (Treasurer)*, [IEEE Chapter at UIUC](#).
- *Publication: 3D shape acquisition via computer vision* — Intl. Sunflower Conf., 2023 ([Scholar](#)).
- Undergraduate Teaching Assistant, CS 450 (Numerical Analysis)

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL

ML & Computer Vision: PyTorch, CUDA, computer vision, perception, NeRF (3D), deep learning

Robotics & Autonomy: ROS 2, motion planning, localization & tracking, state estimation, control systems

Infra & Cloud: Docker, AWS, GPU/CUDA, Slurm HPC, data pipelines, 3D visualization, simulation, CI/CD, Git, Linux